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ENGINEERING
TEXAS A&M UNIVERSITY

Team 64: Sound Localization Robot 4

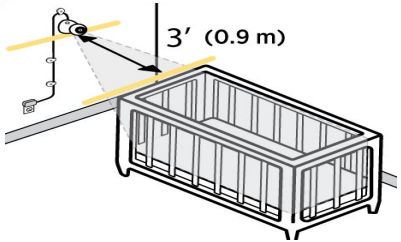
Bi-Weekly Update 1

Samuel Bearden
Cade Christensen
Landon Harris
Xavier Valles
Sponsor: Oscar Moreria
TA: Roman Venegas

Project Summary

Problem Statement:

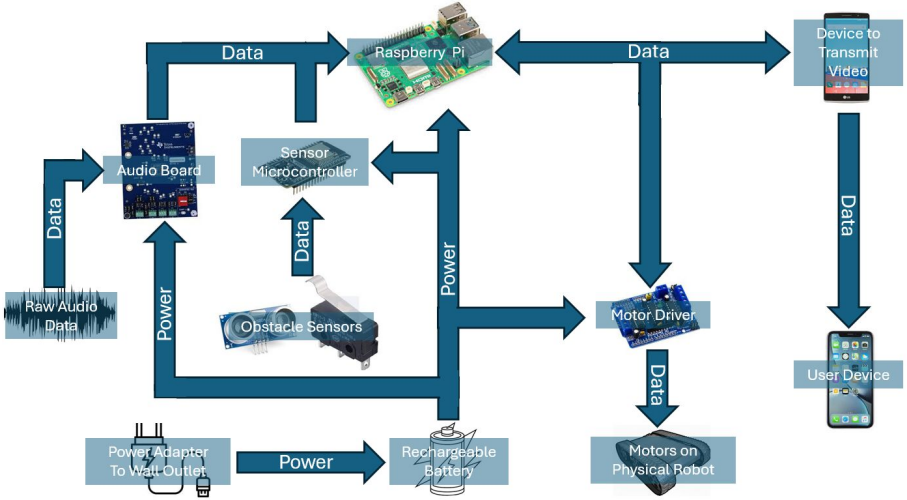
- Current baby monitors are stationary and cover only a limited area
- Some offer camera movement, but still require caretaker control



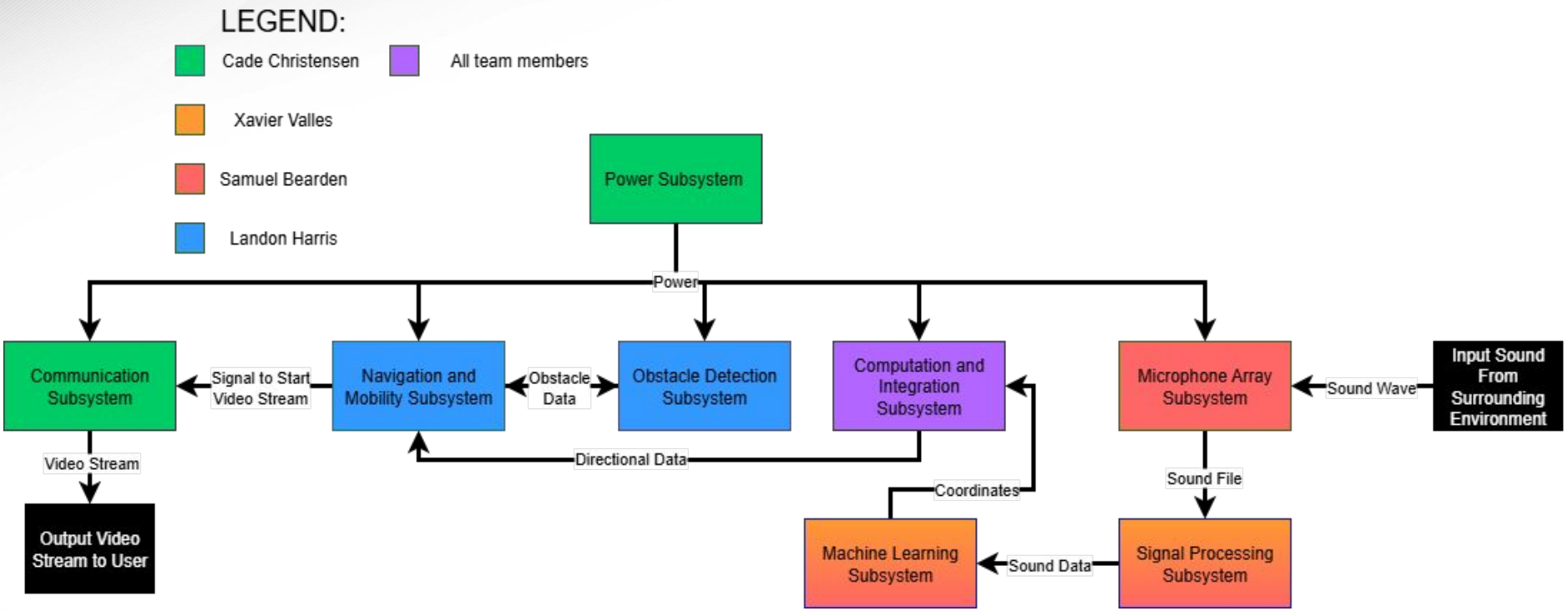
Existing monitors only cover a narrow area

Our Solution:

- A robot that autonomously navigates towards a baby's cry then using sound localization, avoiding obstacles along the way
- Starts a video call to show the baby, giving caretakers peace of mind



System Overview



Major Project Changes for 404

- Minor changes to voltage levels due to more loads
- Sound characterization testing specifications now uses dB instead of sound pressure levels for ambient noise

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Project Timeline

	1/12/2026	1/19/2026	1/26/2026	2/2/2026	2/9/2026	2/16/2026	2/23/2026	3/2/2026	3/9/2026	3/16/2026
ML Based Sound Localization	Ensure model is still operational	Integrate the Raspberry Pi with the TI board so that they can communicate with each other	Configure the Raspberry Pi to record audio using the microphone array	Record new training data using the parabolic dishes	Retrain the model with the new training data	Test model with the whole robot and ensure the robot navigates effectivly	Optimize model and robot path	Optimize model and robot path	Finalize model for robot	
Microphone Array	Redesign Wiring Set up of the microphone array	Receive New Parabolic Dishes	Add Parabolic Dishes and microphones onto the robot	Verify integrity of array on robot						
ML Baby Cry Detection	Review model structure and performance	Get Raspberry Pi and TI board to communicate	Test model using Raspberry PI and TI board	Test autonomous detection	Debug autonomous issues	Full subsytem integration	Resolve any integration issues	Ensure all Validations are met	Prepare for demo	
Noise Filtering	Review Filtering Pipeline technique and performance	Modify dual channel voting code	Test dual channel voting	Integrate with Sound Localization	Debug any integration issues and test up to 65 dB	Full subsytem integration	Resolve any integration issues	Ensure all Validations are met	Prepare for demo	
3D Prints	3D model for parabolic dish sent to modeled for Sound Localization to test	3D model to be finalized and printed	Resdesign to be more modular		Redesign to be printed and tested	Housing for boards to be designed and printed	Integrate and Validate Boards	Final System Test		
Power Subsystem	Begin work on new PCB			Send new PCB to be printed	Test and Validate Boards	Test with Entire Load on the Robot				
Mobile App	Put app onto Raspberry Pi and configure it	Design notification system	Integrate with Sound Characterization for confidence level	Test and Validate	Final System Test					
Sensor Integration	Brainstorm and code new obstacle avoidance algorithm	Mount new sensors onto the robot and rewire sensors with ESP32	Test and debug obstacle avoidance algorithm	Test edge cases for obstacle avoidance algorithm	Code and Integrate ESP32 with Raspberry Pi for sound localization algorithm implementation	Test robot obstacle avoidance algorithm with sound localization algorithm	Fix any issues and Integrate ESP32 with mobile app to communicate once arrives at the baby	Fix any issues and test full system	Finalize validation	Prepare for demo
Motor Control Integration		Perfboard and rewire motors with ESP32								

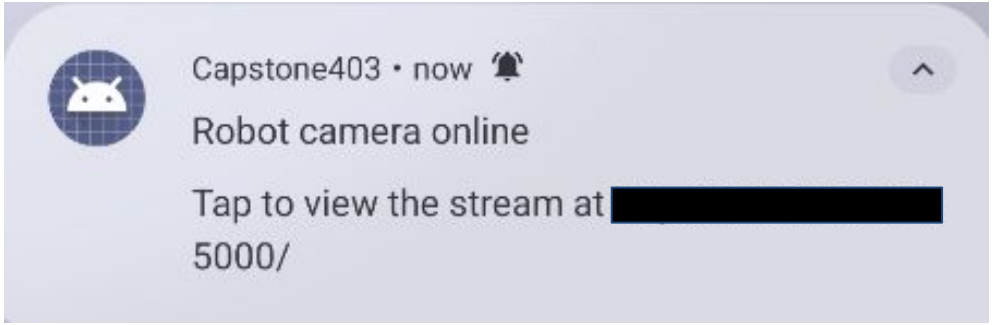
Power and Communication

Cade Christensen

Accomplishments since 403 18 hrs of effort	Ongoing progress/problems and plans until the next presentation
Parabolic dishes have been designed and printed Notification system for the app has been completed	Make the dishes more modular Integrate the app to give the confidence level for the notification Work on PCB to use the battery to power the subsystems.

Power and Communication

Cade Christensen



Sound Characterization & Noise Filtering

Xavier Valles

Accomplishments since 403 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
Hybrid CNN-Transformer model with FP = 1.82% and TP = 99.23% 7-Stage Noise Filtering Pipeline that removes background noise and segments cry audio	Test dual channel voting to help detection and filtering accuracy Test model up to 65 dB using audio board and Raspberry Pi 5

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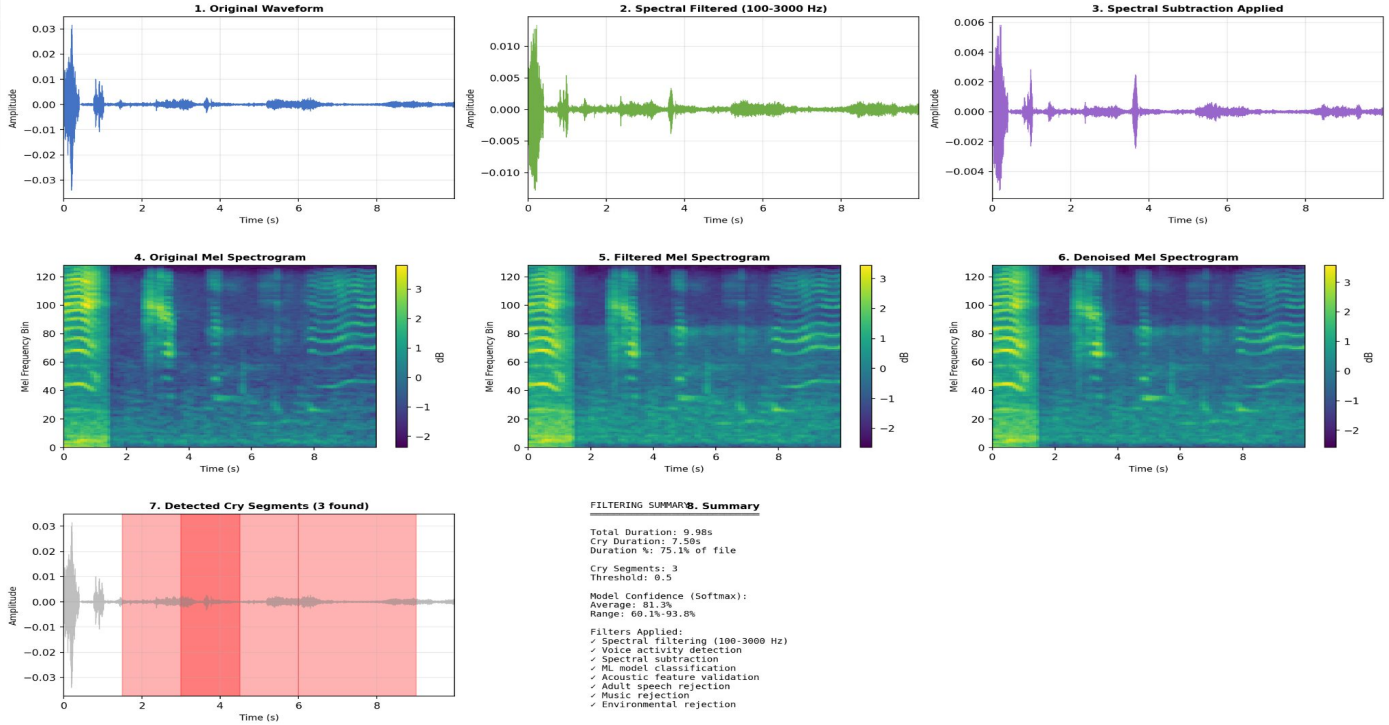
Sound Characterization & Noise Filtering

Xavier Valles



Sound Localization Algorithm & Microphone Array

Samuel Bearden

Accomplishments since 403 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Integrated TI board with Raspberry Pi, allowing me to program TI board from the Pi • Wrote script to set all of the parameters for the Ti board from the Raspberry Pi • Optimized wiring for the microphone array	• Recording audio from the TI board using the Raspberry Pi • Rebuilding microphone array on the actual robot • Recapturing training data with parabolic dishes for my model

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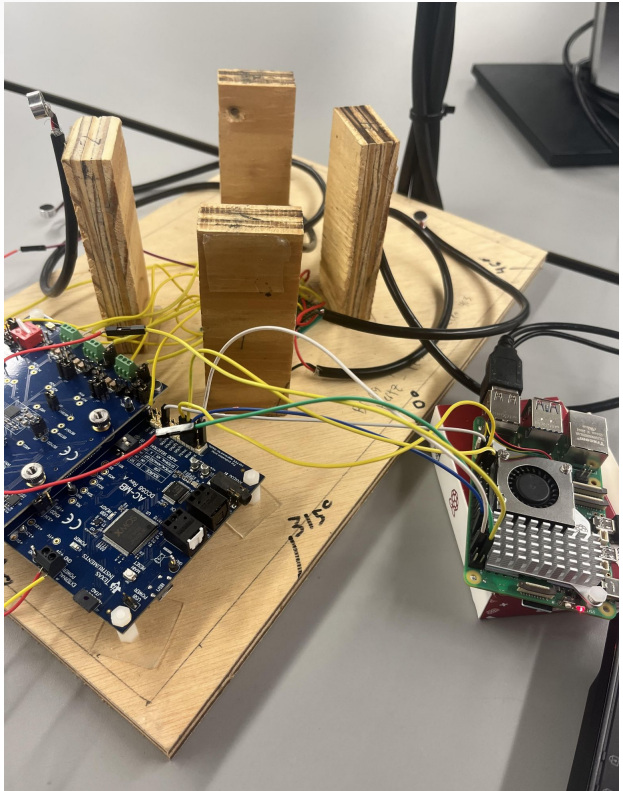
Sound Localization Algorithm & Microphone Array

Samuel Bearden



Navigation/Mobility, & Obstacle Avoidance

Landon Harris

Accomplishments since 403 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Coded obstacle avoidance algorithm and handled edge cases • Perfboarded more connections for ESP32 and level shifter • Rewired entire robot with 3 new ultrasonic sensors	• Test and debug obstacle avoidance algorithm • Integrate ESP32 with Raspberry Pi for sound localization algorithm implementation

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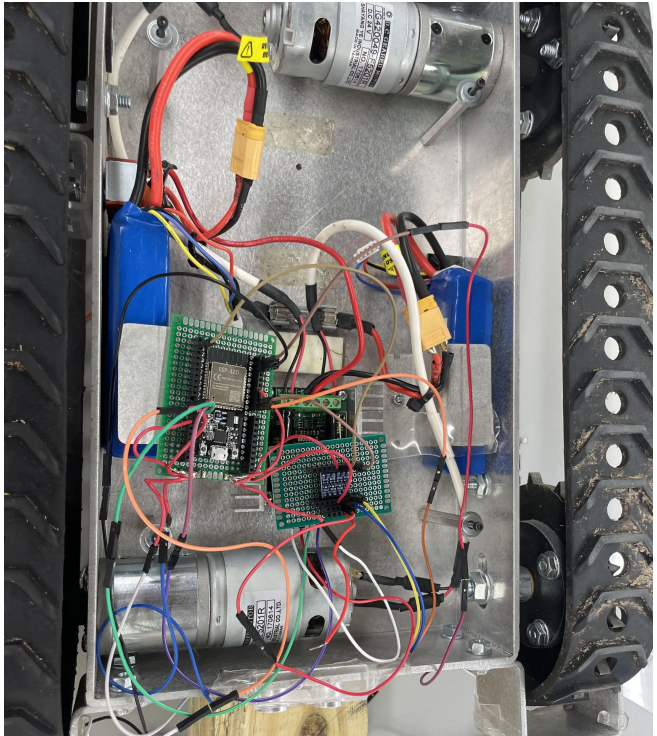
Navigation/Mobility, & Obstacle Avoidance

Landon Harris



Parts Ordering Status

- New PCB for powering the system needs to be ordered once finalized, with the option to use a commercial converter instead if needed
- All other parts needed have been received

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Execution & Plan

Paragraph #	Test Name	Success Criteria	Status	Responsible Engineer(s)
3.2.1.1.	Cry Detection Accuracy	The system correctly identifies an infant cry at least 90% of the time under typical household noise conditions. False positives remain below 10%	Success	Xavier Valles
3.2.1.2.	Sound Localization Accuracy	The system correctly localizes the sound source (room or zone) at least 90% of the time in typical indoor conditions.	Success	Samuel Bearden
3.2.1.3.	Sound Localization Range	The system correctly localizes the sound source (room or zone) from at least 10 feet away from the source	Success	Samuel Bearden
3.2.1.4.	Navigation Response Time	The system autonomously navigates to the identified room within 3 minutes of cry detection, accounting for obstacles and typical indoor conditions	Untested	Landon Harris/Samuel Bearden
3.2.1.5	Obstacle Detection Accuracy	The system detects obstacles in its path with at least 95% accuracy within a detection range of up to 2 meters.	Success	Landon Harris
3.2.3.1.1	Power Consumption	The system operates without exceeding 60 W peak power under normal operation.	Untested	Cade Christensen
3.2.3.1.2	Input Voltage Level	The system operates correctly with input voltage between 22–29 VDC.	Untested	Cade Christensen
3.2.3.2.2	Diagnostic Output	The system correctly diagnoses errors and reports them accordingly	Untested	Samuel Bearden
3.2.4.1	Operating Temperature Range	The system operates reliably within typical household temperature ranges of (68°F to 72°F)	Untested	Cade Christensen
3.2.4.3.	Noise Tolerance	The cry detection system remains effective in background household noise levels up to 65 dB SPL	In Progress	Xavier Valles
N/A	Full System Demo		Untested	Full Team

Thank You For Listening!